

CSE318 Assignment 02

Solving the MAX-CUT Problem using GRASP

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1. Introduction

The MAX-CUT problem is a classic NP-hard optimization problem. It involves partitioning the vertices of a weighted undirected graph into two subsets such that the total weight of the edges crossing the partition is maximized.

2. Algorithm Descriptions

2.1 Randomized Algorithm

This is the simplest approach where each vertex is randomly assigned to one of two partitions with equal probability. Multiple trials are performed, and the average result is taken to mitigate randomness.

Key Characteristics:

- Extremely fast but produces low-quality solutions
- Provides a baseline for comparison
- No guarantee of solution quality

High-level Description: Given a graph $G = (V, E)$ with weights w_{uv} on edges, each vertex $v \in V$ is randomly assigned to one of the two partitions X or Y with equal probability. The total weight of the cut (X, Y) is:

$$w(X, Y) = \sum_{u \in X, v \in Y} w_{uv}$$

This process is repeated for a number of iterations, and the average result is recorded.

Pseudocode:

```
Input: Graph G = (V, E)
Initialize sets A = {}, B = {}

for each vertex v in V:
    Randomly assign v to A or B

return (A, B)
```

Time Complexity: For each iteration, every vertex is assigned randomly, and the cut value is computed by checking all edges. If repeated n times, the total complexity is $O(n(V + E))$. For a single run, it is $O(V + E)$.

2.2 Greedy Algorithm

This deterministic approach builds the solution by always making the locally optimal choice at each step. Vertices are assigned to the partition that maximizes their immediate contribution to the cut.

Key Characteristics:

- Fast execution with reasonable solutions
- Prone to getting stuck in local optima
- No randomness involved

High-level Description: We initialize partitions X and Y using the endpoints of the heaviest edge. For each unassigned vertex $z \in V$, we compute:

$$w_X(z) = \sum_{y \in Y} w_{zy}, \quad w_Y(z) = \sum_{x \in X} w_{zx}$$

Then we assign z to the partition where it contributes the most to the cut.

Pseudocode:

```

Input: Graph  $G = (V, E)$ 
 $u, w$  = endpoints of the heaviest edge in  $G$ 
Initialize sets  $A = \{u\}$ ,  $B = \{w\}$ 

for each vertex  $v$  in  $V$ :
    Calculate edges from  $v$  to  $A$  (EA), and to  $B$  (EB)
    if  $EB > EA$ :
        Add  $v$  to  $A$ 
    else:
        Add  $v$  to  $B$ 

return  $(A, B)$ 

```

Time Complexity: Each vertex is considered once, and for each, the sum of edge weights to both sets is computed. Over all vertices, this totals $O(V + E)$.

2.3 Semi-Greedy Algorithm

A hybrid approach that combines greedy selection with controlled randomness. It maintains a restricted candidate list (RCL) of good options and randomly selects from them, balancing exploration and exploitation.

Key Characteristics:

- Better solution quality than pure greedy
- Tunable via α parameter (0 = pure greedy, 1 = pure random)

- More computationally intensive than greedy

High-level Description: Let $V' = V \setminus (X \cup Y)$ be the set of unassigned vertices. For each $v \in V'$, compute the greedy score:

$$\sigma_X(v) = \sum_{u \in X} w_{vu}, \quad \sigma_Y(v) = \sum_{u \in Y} w_{vu}$$

$$g(v) = \max\{\sigma_X(v), \sigma_Y(v)\}$$

Compute:

$$w_{\min} = \min_{v \in V'} g(v), \quad w_{\max} = \max_{v \in V'} g(v)$$

$$\mu = w_{\min} + \alpha(w_{\max} - w_{\min})$$

The restricted candidate list (RCL) consists of vertices with $g(v) \geq \mu$. A vertex is randomly chosen from the RCL and assigned to the appropriate partition.

Pseudocode:

```

Input: Graph G = (V, E)
u, w = endpoints of the heaviest edge in G
Initialize sets A = {u}, B = {w}
Initialize unvisited = V

while unvisited not empty:
    For each v in unvisited:
        compute gain if added to A or B

    Build Restricted Candidate List (RCL) of top-k gain vertices
    Randomly select a vertex v from RCL
    Assign v to the set with higher gain
    Remove v from unvisited

return (A, B)

```

Time Complexity: For each of V assignments, all unassigned vertices are evaluated, and for each, edge sums are computed. This results in $O(V(V + E))$ overall.

2.4 Local Search

Starts with an initial solution and iteratively improves it by moving vertices between partitions if doing so increases the cut weight. Continues until no further improvements can be made.

Key Characteristics:

- Can significantly improve existing solutions
- Performance depends heavily on initial solution quality
- May get stuck in local optima

High-level Description: Given an initial cut $(S, \bar{S})$, for each vertex v , the change in cut value by moving it to the opposite partition is given by:

$$\delta(v) = \begin{cases} \sigma_S(v) - \sigma_{\bar{S}}(v), & \text{if } v \in S \\ \sigma_{\bar{S}}(v) - \sigma_S(v), & \text{if } v \in \bar{S} \end{cases}$$

We iteratively move the vertex with the best positive $\delta(v)$ until no improvement is possible (local optimum reached).

Pseudocode:

Input: Graph $G = (V, E)$

Start with initial cut (A, B) , e.g., random or greedy

```
repeat
    improvement = false
    for each vertex v:
        Calculate gain if moved to opposite set
        if gain > 0:
            Move v to other set
            improvement = true
until improvement == false

return (A, B)
```

Time Complexity: Each iteration checks all vertices and their neighbors, which is $O(V + E)$ per iteration. If k is the number of improvement steps, total complexity is $O(k(V + E))$.

2.5 GRASP (Greedy Randomized Adaptive Search Procedure)

A metaheuristic that combines semi-greedy construction with local search. Multiple iterations are performed, each generating a new solution which is then locally optimized.

Key Characteristics:

- Combines exploration (semi-greedy) and exploitation (local search)
- Typically produces the highest quality solutions
- More computationally intensive than other methods

High-level Description: This method performs multiple iterations. In each iteration:

- A solution is constructed using the Semi-Greedy algorithm
- The solution is improved using Local Search

Among all iterations, the best cut value found is selected as the final result.

Pseudocode:

Input: Graph $G = (V, E)$, MaxIterations

```
best_cut = null
best_value = -inf
```

```

for i = 1 to MaxIterations:
    (A, B) = SemiGreedyConstruction(G)
    (A, B) = LocalSearch(G, A, B)
    value = CutValue(G, A, B)

    if value > best_value:
        best_cut = (A, B)
        best_value = value

return best_cut

```

Time Complexity: For n iterations, each performing semi-greedy construction ($O(V(V+E))$) and local search ($O(k(V+E))$), the total is $O(n(V(V+E) + k(V+E)))$. For practical purposes, this is often written as $O(n(V+E + k(V+E)))$.

Note on Parameters:

Alpha Parameter Tuning: The semi-greedy algorithm is tunable via the α (alpha) parameter, which controls the balance between greediness and randomness when selecting the next vertex to assign:

- **When $\alpha = 0$:** The threshold μ becomes the minimum possible value ($w_{\min}$). Only the most "greedy" candidates (those with the highest gain) are chosen, making the algorithm behave like a pure greedy algorithm.
- **When $\alpha = 1$:** The threshold μ becomes the maximum possible value ($w_{\max}$). All candidates with the highest possible gain are included, but since the restricted candidate list can be large, the selection among them is random, making the algorithm behave more randomly.
- **For $0 < \alpha < 1$:** The threshold μ is between $w_{\min}$ and $w_{\max}$. The restricted candidate list includes all candidates whose greedy value is at least μ , so the algorithm mixes greedy selection with random choice among good candidates.

Detailed Example: In the semi-greedy algorithm, for each unassigned vertex, you calculate a "greedy value" (how good it would be to assign it to a set). You find the minimum ($w_{\min}$) and maximum ($w_{\max}$) of these values, then set a threshold: $\mu = w_{\min} + \alpha \times (w_{\max} - w_{\min})$.

When $\alpha = 1$: $\mu = w_{\min} + 1 \times (w_{\max} - w_{\min}) = w_{\max}$. So, only vertices whose greedy value is at least $w_{\max}$ are considered for assignment. Usually, only the vertices with the highest possible greedy value are included in the restricted candidate list. If multiple vertices have this highest value, you pick one randomly from them.

This introduces randomness into the selection, even though you're only considering the best candidates. With $\alpha = 1$, you always pick from the best candidates, but if there are ties, you choose randomly among them, making the process less deterministic and more random than pure greedy.

Summary:

- $\alpha = 0$: Pure greedy (always picks the best candidate)
- $\alpha = 1$: Pure random among the best candidates
- $0 < \alpha < 1$: Mix of greedy and random, allowing you to tune the balance between exploitation (greedy) and exploration (random)

- α ($0 \leq \alpha \leq 1$): Controls how greedy the semi-greedy construction is. $\alpha = 0$ is fully greedy, $\alpha = 1$ is fully random. In practice, α is often set to 0.5 to balance exploration and exploitation.
- **MaxIterations**: Number of iterations GRASP will run. Higher values increase quality but require more time. It is often set to a small number (e.g., 50) for practical purposes.

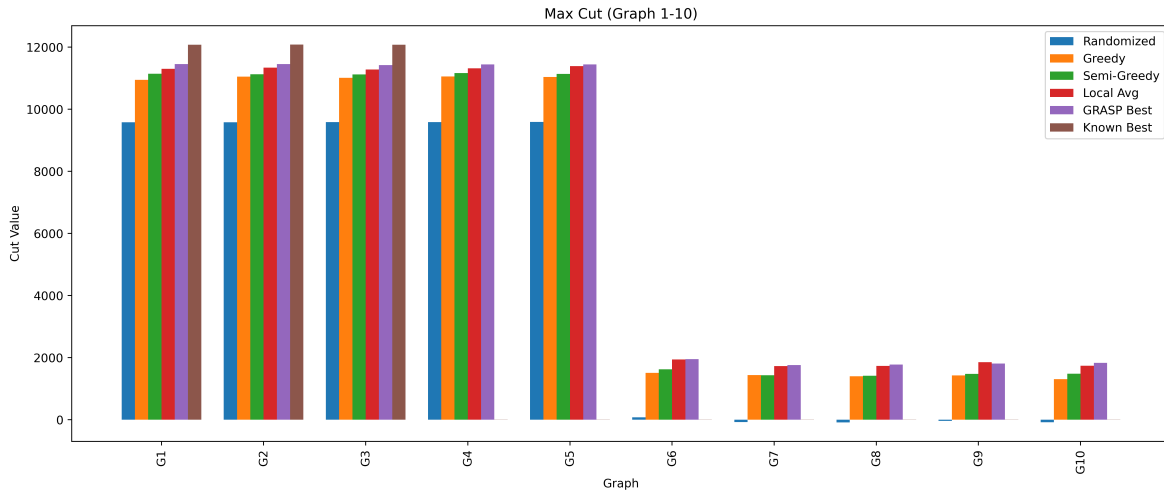
3. Algorithm Comparison

- **GRASP** consistently yields the highest cut weights across all benchmark graphs due to its balanced use of exploration (semi-greedy) and exploitation (local search).
- **Local Search** is very effective when combined with a good initial solution, significantly improving cut values.
- **Semi-Greedy** generally performs better than Greedy, as it avoids falling into poor local optima by introducing controlled randomness.
- **Greedy** is the fastest method and provides reasonable results, but may miss better solutions due to its deterministic nature.
- **Randomized** gives diverse but typically suboptimal results and is mostly useful as a baseline.

Table 1: Algorithm Comparison

Algorithm	Quality	Speed	Deterministic	Complexity
Randomized	Low	Very Fast	No	$O(n(V + E))$
Greedy	Medium	Fast	Yes	$O(V + E)$
Semi-Greedy	Medium-High	Medium	No	$O(V(V + E))$
Local Search	High	Slow	Depends on initial	$O(k(V + E))$
GRASP	Very High	Slowest	No	$O(n(V(V + E) + k(V + E)))$

4. Experimental Results



5. Additional Notes and Observations

- In most graphs, Greedy and Semi-Greedy produce close results, but Semi-Greedy slightly edges out due to randomization.
- Local search greatly improves solutions when initialized from Greedy or Semi-Greedy.
- GRASP consistently improves upon both, especially in denser graphs.
- Randomized solutions vary greatly; useful for baseline or stress-testing other heuristics.
- Execution time scales with graph size and number of iterations. GRASP's performance/quality tradeoff is tunable.

6. Conclusion

GRASP, by combining diversification (through semi-greedy construction) and intensification (through local search), consistently yields high-quality solutions to the MAX-CUT problem. While more computationally intensive than simpler approaches, it provides the best trade-off between solution quality and runtime for many practical applications. The experimental results demonstrate that the additional computational effort is justified by the significant improvements in solution quality.